

Task 6 — Service Validation

Each CampusEats service is checked against the five properties of a web service:

1. Reachable over a network
2. Self-contained
3. Has a contract
4. Independent
5. Loosely coupled

Validation Table

Service	Reachable	Self-contained	Has Contract	Independent	Loosely Coupled
Accounts Service	✓	✓	✓	✓	✓
Catalogue Service	✓	✓	✓	✓	✓
Orders Service	✓	✓	✓	✓	✓
Payments Service	✓	✓	✓	✓	✓
Delivery Service	✓	✓	✓	✓	✓
Notifications Service	✓	✓	✓	✓	✓

Accounts Service

Property	Result	Reason
Reachable over a network	✓	Other services can call its published operations over the network.
Self-contained	✓	It manages user profiles, addresses, and login responsibilities.
Has a contract	✓	Operations such as <code>getAddress()</code> and <code>getProfile()</code> define its interface.
Independent	✓	Its internal implementation can change without requiring changes from callers.
Loosely coupled	✓	Other services use its published contracts instead of accessing its internal data directly.

Catalogue Service

Property	Result	Reason
Reachable over a network	✓	Other services can call catalogue operations over the network.
Self-contained	✓	It manages restaurants, menus, prices, and item availability.
Has a contract	✓	<code>listRestaurants()</code> , <code>getMenu()</code> , and <code>checkItem()</code> define its contract.
Independent	✓	Its internal implementation can change without requiring changes to Orders.
Loosely coupled	✓	Orders uses <code>checkItem()</code> instead of directly accessing catalogue data.

Orders Service

Property	Result	Reason
Reachable over a network	✓	Students and other services can access its published operations.
Self-contained	✓	It manages carts, orders, and order status.
Has a contract	✓	<code>addToCart()</code> , <code>placeOrder()</code> , <code>getOrder()</code> , and <code>cancelOrder()</code> define its contract.
Independent	✓	Its internal order implementation can change without exposing its internal data.
Loosely coupled	✓	It communicates with other services through their published contracts.

Payments Service

Property	Result	Reason
Reachable over a network	✓	Orders can call payment operations over the network.
Self-contained	✓	It handles payment and refund responsibilities.
Has a contract	✓	<code>charge()</code> and <code>refund()</code> define its public contract.
Independent	✓	The payment provider can be changed internally without changing Orders.
Loosely coupled	✓	Orders receives payment results through the payment contract.

Delivery Service

Property	Result	Reason
Reachable over a network	✓	Orders can call delivery operations over the network.
Self-contained	✓	It manages riders, assignments, and delivery tracking.
Has a contract	✓	<code>assignRider()</code> and <code>getTracking()</code> define its public operations.
Independent	✓	Its rider assignment and tracking implementation can change internally.
Loosely coupled	✓	Orders requests delivery through the published contract.

Notifications Service

Property	Result	Reason
Reachable over a network	✓	Other services can request notifications over the network.
Self-contained	✓	It handles notification delivery.
Has a contract	✓	<code>send()</code> defines the operation available to other services.
Independent	✓	Its notification provider or internal implementation can change without changing callers.
Loosely coupled	✓	Other services call <code>send()</code> rather than accessing its internal data directly.